

Saranpat Prasertthum

(909) 462 5694 | Saranpatp.pr@gmail.com | linkedin.com/in/saranpatp

WORK EXPERIENCE

Overhill Farms

Vernon, CA

Data Scientist, Full-time

July 2024 – Current

- Saved \$2.3M annually by reducing warehouses location using clustering algorithms, leading a strategic warehouse consolidation project, and optimizing shipping routes while maintaining capacity and efficiency.
- Decreased promotion spending by \$300K while increasing sales by building a data warehouse from SAP to project promotion ROI and leveraging IRI POS data to optimize spending.

Metrixx LLC

Chicago, IL

Quantitative Developer, Full-time

May 2023 – July 2024

- Deployed an asset recommendation system that uncovered profitable opportunities for clients by scanning 1,000+ stocks, ETFs, and futures using real-time social media sentiment analysis and generative AI for analysis.

Tower Research Capital

Chicago, IL

Practicum Project Researcher, Part-time

January – May 2023

- Prevented over \$30M in annualized damages and achieved a 97% R-score by identifying and utilizing the most reliable price feeds from medium to small crypto exchanges, enhancing trade stability during market fluctuations.
- Back testing of pricing across 100+ cryptocurrency trading pairs by designing a high-performance Go module that simulates millisecond-level price movements and emulates exchange activity with precision.

Blockfint

Bangkok, Thailand

Software Engineer, Full-time

January 2020 – September 2021

- Generated over \$500K in bond sales and seamlessly supported 10K users on launch day by pioneering the development of Thailand's first digital bond trading platform, engineered for scalability and user experience.
- Tracked over \$1 million in agricultural produce and prevented illegally imported palm oil products without chemical alteration by engineering a Python back-end for a tracking system that verified product legitimacy using blockchain.

TECHNICAL SKILLS

- **Programming Languages:** Python (5+ years) , SQL (5+ years) , C++ (4+ years), Go-lang (3+ years), C, JavaScript, Java, GraphQL, R, bash, APIs
- **Tools:** Bloomberg, AWS, Google Cloud, Digital Ocean, Docker, Git, Linux, JIRA, Trello, MS Teams, Jenkins, Tableau
- **Frameworks:** Flutter, FastAPI, Gin, Flask, Django, MongoDB, Vue.js
- **Financial Skills:** Greeks, Delta-hedging, Portfolio Optimization, Valuation, Asset Pricing (Fixed-Income, Exotic Options, Swaptions), Interest Rate Models (Vasicek, Black-Derman-Toy, Hull-White), Monte Carlo

EDUCATION

University of Illinois Urbana Champaign

Urbana-Champaign, IL

Master of Science in Financial Engineering GPA 3.75/4.00

August 2022 – December 2023

Courses: Stochastic Calculus, High-Frequency Trading, Option Mkt Making, Term-structure, Algo Trading System, Financial Derivatives, Numerical Methods, Statistics, Machine Learning, Market Micro-structure, Term Structure

Chulalongkorn University

Bangkok, Thailand

Bachelor of Science in Computer Engineering

August 2016 – May 2020

Courses: Software Engineering, Computer Security, Data Science, Neural Network, Pattern Recognition

PROJECT HIGHLIGHT

High Frequency Network Route Optimization

- Optimized order placement by analyzing exchange fiber-optic networks and mitigating latency from natural disruptions, achieving an 80% rate of on-exchange fills (Futures market).

Option Trading Strategy (Bloomberg)

- Implemented a Dynamic-hedging algorithm to manage options portfolio risk. Reduced delta exposure by 30% through automated adjustments of underlying asset positions, leading to improved portfolio stability and targeted returns.

Momentum Base Strategy on Commodities (Bloomberg)

- Engineered a rank-reversal momentum strategy leveraging prior period losers and winners for trade selection. Incorporated transaction cost modeling for real-world applicability, achieving a 30% outperformance compared to the research paper's baseline.
- Extracted 20 years of 25 futures contracts data using the Bloomberg Python API, streamlining the backtesting process.

Term Structure Models Calibration and Exotic Pricing (Python: Numpy, Pandas, Scipy)

- Leveraged Bloomberg's caplet volatility surfaces and yield curve data to derive accurate exotic pricing for Bermudan swaptions, barrier options, and other structured products.
- Calibrated one-factor BDT and HW short-rate models in Python using caplet and swaption yield curve stripping, resulting in less than 3% and 18% absolute error, respectively, compared to the short rates from the Bloomberg terminal.

Market Making Strategy on Real Market Data (C++)

- Analyzed real market data to strategically uncover hidden and iceberg orders, providing a crucial edge that substantially elevated market-making strategy. Performed Backtest on RCM Strategy Studio using C++ yielded an approximately 53% win rate, across the 10 most actively traded cryptocurrency pairs.